

SUBJECT 2 – QUANTUM OPTIMIZATION

MODULE 1: Fundamentals of Optimization

Introduction to Optimization - Objective Functions - Decision Variables - Constraints - Feasible and Infeasible Solutions - Constrained and Unconstrained Optimization - Binary Optimization - Combinatorial Optimization - NP-Hard Problems

MODULE 2: Quantum Computing Fundamentals for Optimization

Introduction to Quantum Computing - Qubits - Quantum States - Superposition - Entanglement - Quantum Gates - Quantum Circuits - Measurement - Quantum Computing for Optimization

MODULE 3: QUBO and Ising Model Formulations

Binary Decision Variables - Quadratic Unconstrained Binary Optimization (QUBO) - QUBO Objective Function - QUBO Matrix - QUBO Formulation - Ising Model - Spin Variables - QUBO to Ising Transformation - Energy Minimization

MODULE 4: Quantum Optimization Algorithms

Introduction to Quantum Optimization Algorithms - Variational Quantum Algorithms - Quantum Approximate Optimization Algorithm (QAOA) - Variational Quantum Eigensolver (VQE) - CVaR-VQE - Cost Hamiltonian - Mixer Hamiltonian - Parameterized Quantum Circuits - Classical Optimization of Parameters

MODULE 5: Constraint Handling in Quantum Optimization

Types of Constraints - Equality Constraints - Inequality Constraints - Constraint Encoding - Penalty Methods - Penalty Coefficients - Slack Variables - Binary Encoding of Slack Variables - Qubit Overhead - Feasibility and Optimality

MODULE 6: Lagrangian Relaxation and Slack-Free Optimization

Lagrangian Relaxation - Lagrange Multipliers - Lagrangian Function - Dual Problem - Lagrangian Duality - Constraint Relaxation - Subgradient Method - Dual Averaging - Bundle Method - Cutting Plane Method - Augmented Lagrangian - Slack-Free Formulations - Unbalanced Penalty Methods

MODULE 7: Evaluation of Quantum Optimization

Solution Quality - Feasibility - Optimality Gap - Variable Reduction - Qubit Reduction - Circuit Complexity - Computational Cost - Comparison of Optimization Approaches